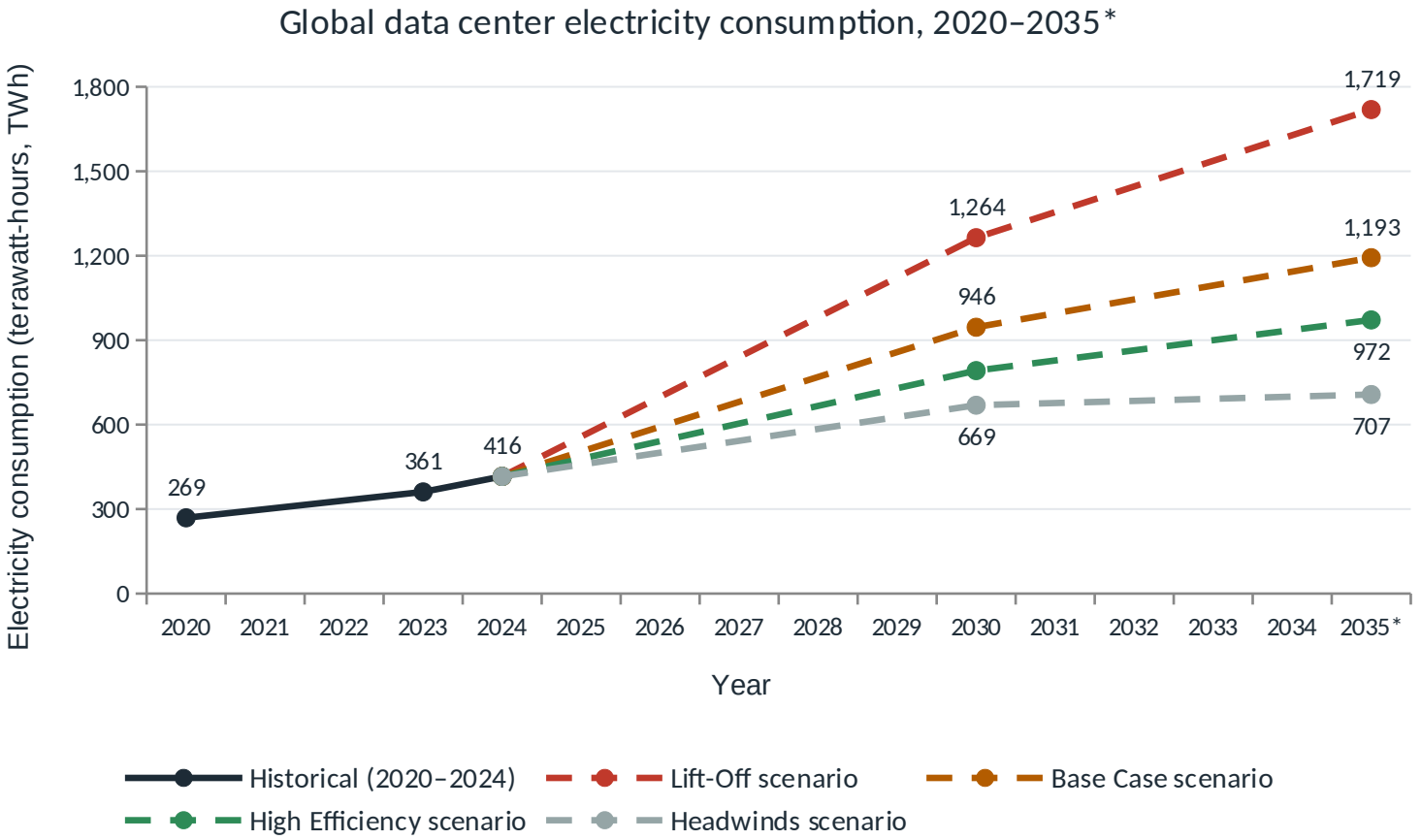
Prepared by Will Louiselle | October 2026 | For internal discussion

AI is driving global data center electricity use to more than double by 2030



Source data: IEA, Energy and AI data annex (updated June 16, 2025), World Data sheet: https://www.iea.org/data-and-statistics/data-product/energy-and-ai

*2035 values are exploratory scenarios. Dashed lines are IEA projections, not measurements.

Key metrics

2024 consumption (IEA estimate)	**416 TWh**
2030 projection, Base Case	**946 TWh**
Change, 2024 to 2030	**+127% (2.3×)**
2035 range across scenarios	**707–1,719 TWh**

Key findings

- Consumption rose 55% from 2020 to 2024, from 269 to 416 TWh
- AI-optimized servers drive nearly half of the projected increase
- 2030 Base Case demand is roughly equal to Japan's total electricity use today

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Efficiency gains slow growth but do not reverse it: even the High Efficiency case more than doubles 2024 demand by 2035

Scenario tracker: global data center electricity (TWh)

Scenario	Main driver	2030	2035*	2035 vs. 2024	2035 vs. Base
Lift-Off	Faster AI adoption	1,264	1,719	4.1×	+44.1%
Base Case	Current trajectory	946	1,193	2.9×	—
High Efficiency	**Stronger efficiency gains**	**792**	**972**	**2.3×**	**−18.5%**
Headwinds	Supply-chain and grid bottlenecks	669	707	1.7×	−40.7%

2024 baseline: 416 TWh (IEA estimate). *2035 values are exploratory scenarios.

Mitigation levers and evidence

Lever	Evidence
Model efficiency	Google cut energy per AI prompt 33× in one year (May 2024–May 2025) with leaner model designs
Facility efficiency	2024 overhead per kWh of IT load: 0.14 kWh at hyperscale sites vs. 0.92 kWh at enterprise sites
Clean, flexible power	Tech sector signed ~40% of corporate renewable PPAs in 2025; on-site batteries add grid flexibility

DATA LIMITATION

The IEA dataset measures electricity for all data center workloads. It does not isolate AI's share, and it does not measure water use or the emissions from manufacturing chips.

BUSINESS IMPLICATION

Build energy into AI investment decisions: require vendors to report efficiency metrics (PUE, energy per query, clean-power share), prioritize efficient cloud capacity over on-premises servers, and default to the smallest model that meets the need.

Sources: International Energy Agency (2025), Energy and AI data annex, https://www.iea.org/data-and-statistics/data-product/energy-and-ai; IEA (2025), Energy and AI, https://www.iea.org/reports/energy-and-ai; IEA (2026), Key Questions on Energy and AI, https://www.iea.org/reports/key-questions-on-energy-and-ai; Google (2025), Measuring the Environmental Impact of Delivering AI at Google Scale, https://arxiv.org/pdf/2508.15734